

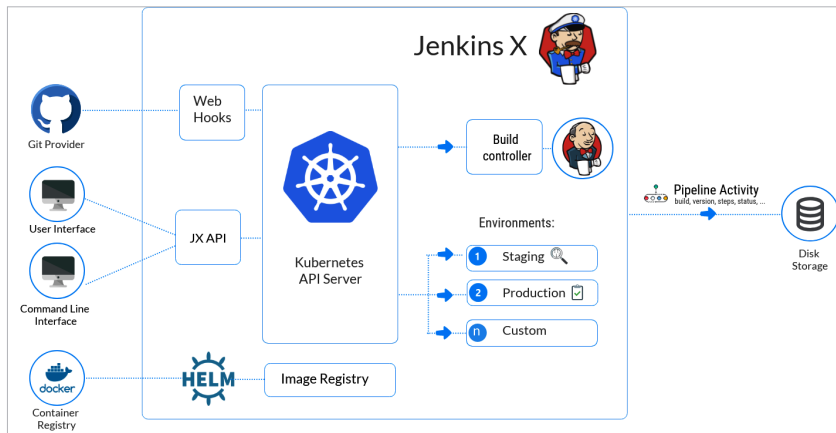


Figure 1: Kubernetes-based architecture for deploying microservices in a scalable and repeatable manner

(e.g., dev, staging, production). Helm and Draft are used for configuration management, utilising charts and templates to simplify application deployment in Kubernetes. Jenkins X introduces a ‘serverless Jenkins’ configuration, allowing deployment to a Kubernetes cluster without Jenkins. The ‘static Jenkins’ architecture is deprecated, and Jenkins X exclusively supports the Tekton-based configuration, with users urged to migrate for continued updates and support.

Tools and frameworks

Jenkins X incorporates several tools and frameworks. The latest versions integrate and utilise:

- **Jenkins**, the core CI/CD engine automating software delivery, managing build, test, and deployment with extensive plugin support.
- **Kubernetes**, the orchestration platform that automates the deployment, scaling, and administration of containerised applications in a distributed environment.
- **Helm**, the Kubernetes package manager simplifying installation, upgrade, and management of applications using pre-configured resource packages (charts).
- **Tekton**, the Kubernetes-native framework facilitating CI/CD

pipeline creation with container-based workflows for granular control and flexibility.

- **GitOps**, the methodology that leverages Git repositories as a source of truth for configuration and deployment management, ensuring system state aligns with desired configurations stored in Git.
- **Docker**, the platform for containerisation, enabling packaging applications and dependencies into standardised units (containers) for consistency and portability.
- **JX CLI**, the Jenkins X command-line interface streamlining management of installations, applications, pipelines, and configurations in Jenkins X via command line.

These tools work together to automate the software development life cycle, offering a streamlined process for building, testing, deploying, and managing applications in Kubernetes-based environments.

Jenkins X streamlines cloud-native CI/CD on Kubernetes, yet proves beneficial locally during initial development and testing. Local deployment allows rapid iteration and testing before pushing code to cloud-based clusters, facilitating quicker debugging. It also serves as a valuable learning tool, providing a controlled environment for teams to grasp Jenkins

X, GitOps, and Kubernetes concepts, easing the transition to a cloud-native setup. Additionally, local deployment addresses internal control concerns by keeping sensitive code within the organisation’s network, ensuring compliance with security standards while facilitating efficient collaboration.

Importance of Jenkins X

The importance of Jenkins X lies in its ability to address the unique challenges and requirements of modern, cloud-native software development.

- **Native Kubernetes support:** Jenkins X is purpose-built for Kubernetes, simplifying application building, deployment, and management in containerised, microservices-oriented environments.
- **Automation:** It automates CI/CD pipelines, accelerating development cycles, reducing human errors, and ensuring reliable and consistent deployments.
- **GitOps:** Promoting GitOps practices, Jenkins X versions entire application and environment configurations in Git repositories, enhancing collaboration, traceability, and rollback capabilities.
- **DevOps best practices:** It incorporates DevOps practices, including automated testing, code reviews, and security scanning, ensuring high-quality code and early identification of security vulnerabilities.
- **Cloud-native ecosystem integration:** Jenkins X seamlessly integrates with cloud-native tools like Helm, Scaffold, and Tekton, streamlining application deployment and resource management.
- **Multi-environment support:** It supports multiple environments, enabling easy application promotion from development and testing to production in a Kubernetes setting.
- **Scaling and scalability:** Jenkins X scales to meet growing application and team needs, handling increased